

# Technical Requirement Document

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Here's the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

**Technical Requirement ID: TR-E2E-001**

**Related Functional Requirement(s): FR-E2E-001**

**Objective: Implement a PySpark job to read address change candidate records from an Oracle staging source, generate a flat-file style target dataset, and update the staging table with processing metadata and key identifiers.**

## **Target Cluster Configuration:**

- Cluster Name: E2E Processing Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard\_DS3\_v2
- Driver Node: Standard\_DS3\_v2 (1 node)
- Worker Nodes: Standard\_DS3\_v2 (2-5 nodes)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Oracle JDBC driver, Spark Oracle connector

## **Source Data Details:**

- Dataset Name: STG\_E2E\_AC\_TXDBH\_DATA
- Location: Oracle staging database
- Format: Oracle table
- Description: Address change candidate records
- Dataset Name: T\_YUYU\_CLNT and T\_YUYUK\_CLN
- Location: Oracle database
- Format: Oracle tables
- Description: Lookup tables for owner name components

## **Target Data Details:**

- Output Name: DPAddressChangeYUYU
- Location: Designated file system path
- Format: Flat file (delimited)
- Description: Target dataset containing specific fields

- Output Name: STG\_E2E\_AC\_TXDBH\_DATA\_TGT
- Location: Oracle staging database
- Format: Oracle table
- Description: Updated staging table with processing metadata and key identifiers

### **Job Flow / Pipeline Stages:**

- Stage 1: Source Read - Read address change candidate records from STG\_E2E\_AC\_TXDBH\_DATA
- Stage 2: Pre-Processing - Pass through key and address/phone fields, expose “PPAY\_” fields
- Stage 3: Enrichment - Perform lookups on T\_YUYU\_CLNT and T\_YUYUK\_CLN for owner name components
- Stage 4: Business Transformations - Apply date formatting, postal code normalization, address fallback logic, phone number formatting, and owner name concatenation
- Stage 5: Outputs - Generate DPAddressChangeYUYU flat file and update STG\_E2E\_AC\_TXDBH\_DATA\_TGT

### **Data Transformations / Business Logic:**

- Step: Derive formatted dates
- Description: Apply date formatting business rule (BR-03)
- Transformation Logic: Use Spark SQL date formatting functions
- Step: Normalize postal code
- Description: Remove hyphens from postal code
- Transformation Logic: Use Spark SQL string manipulation functions
- Step: Apply address fallback logic
- Description: Use fallback values when ZIP is missing
- Transformation Logic: Use Spark SQL conditional logic
- Step: Format phone numbers
- Description: Apply phone number formatting business rule (BR-06)
- Transformation Logic: Use Spark SQL string manipulation functions
- Step: Concatenate owner names
- Description: Derive owner name components using lookup tables
- Transformation Logic: Use Spark SQL join and string concatenation functions

### **Error Handling and Logging:**

- Log error messages and counts at each stage
- Use try-except blocks to catch and handle exceptions
- Log job execution metrics (e.g., records processed, time taken)

- Use Databricks logging mechanisms to store logs in a designated location